How big must a Kalshi–Polymarket price gap get before it closes, and how fast does it close?

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Harvard University, AM215 Mini-project, Fall 2026
Code: https://github.com/alexmeyl/Kalshi-Polymarket-gap-modeling

Abstract. Kalshi and Polymarket list the same events, but the two prices for the same outcome rarely agree. Arbitrage should pull them together, yet trading is not free, so small gaps should survive. We model the gap as a *band* process: inside a no-trade band of half-width c the gap wanders freely; beyond it, the excess decays at rate κ. Fitting nine matched markets at one-minute resolution, we find bands of 0–1.5¢ and half-lives from 3.6 minutes (World Series) to 2.9 hours (Swift engagement), each with block-bootstrap 95% ranges. The fitted band tracks the measured Kalshi trading cost across markets (Spearman $\rho = +0.77$). The code reproduces the closed-form AR(1) results in the $c = 0$ limit to within 0.3%, and on the last 30% of each market the model beats a random walk in 7 of 9 markets. Simulations show the estimates are biased in fast, quiet markets, and a test on nine further markets does not confirm that the band carries over; we explain why that test is weak and what would make it fair.

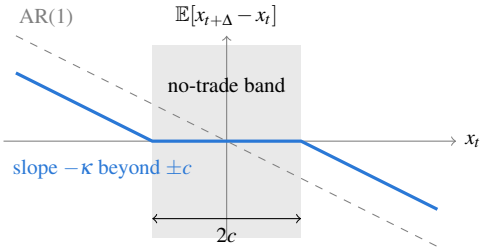


Figure 1. The band rule. Inside $|x| \le c$ there is no pull; beyond it the expected change falls with slope $-\kappa$. Plain AR(1) (dashed) pulls back every gap.

1 Question

A prediction-market price is a probability: a YES contract that trades at 0.62 pays $1 if the event happens, so the market thinks it is 62% likely. When Kalshi and Polymarket list the same event, the two YES prices K_t and P_t should agree, but they often do not. We call their difference the *gap*, $G_t = K_t - P_t$; a gap of 0.01 is one cent.

Why gaps close. Neither platform lets you short-sell, but every market has a NO contract worth about 1 − YES. If YES costs P_A on site A and $P_B > P_A$ on site B, buying YES on A and NO on B costs $P_A + (1 - P_B) = 1 - (P_B - P_A)$, and exactly one of the two contracts pays $1. The profit equals the gap, whatever happens. The trade also removes the gap: buying YES on A lifts A's price, buying NO on B lowers B's YES price.

Why small gaps do not close. With real quotes the trader pays the *ask* on both sites, so the trade only makes money if

$$\text{bid}_B^{\text{YES}} - \text{ask}_A^{\text{YES}} > f_A + f_B, \tag{1}$$

where f are the fees. Measured from mid-prices, a gap must exceed roughly half of each site's spread plus fees. The money is also locked up until the event resolves, and the two sites' rules may settle differently. Below that threshold nobody trades, and the gap simply drifts.

The two numbers we estimate. For each of nine matched markets we ask: (i) how big must the gap be before anything pulls it back, a band c in cents; and (ii) once it is beyond c, how fast does the excess shrink, a half-life $t_{1/2}$ in minutes. Both come directly out of the model in §3.

Why this model. Plain mean reversion (AR(1)) pulls back every gap, even a tenth of a cent, which contradicts (1). A band (threshold) autoregression is the simplest model with a cost-sized dead zone (Figure 1). Its band has a physical meaning we can check against measured costs, it contains AR(1) as the case $c = 0$, which gives an exact verification limit (§5), and it lets us test whether the band adds anything over plain mean reversion (§6).

2 Assumptions

Table 1 lists every assumption, the ones we chose on purpose and the ones hidden in the equations, code or data handling, each with where it breaks in our own data. Three hidden assumptions matter most for the results. *One closing speed*: the rule has a single κ, but in the Super Bowl and World Series gaps close fast at first and then slowly, so $t_{1/2}$ there describes only the first few minutes (§6). *Simultaneous quotes*: we treat prices matched within 30 s as simultaneous, but quiet markets carry stale quotes, which our simulations show pull $\hat{c}$ and $\hat{\kappa}$ down (§5). *Same settlement*: the arbitrage in §1 is risk-free only if both contracts pay on the same outcome; the 2024 election pair did not quite match ("Trump *or another Republican*" vs "Trump"), which may explain its steady 3¢ offset.

3 Model

Rule. Let m be the market's median gap and $x_t = G_t - m$ the gap measured from its usual level. With step $\Delta = 5$ min and the *excess* beyond the band

$$e_c(x) = \text{sign}(x)\,\max\big(0,\, |x| - c\big), \tag{2}$$

Table 1. Assumptions. "Chosen": a deliberate modeling choice. "Hidden": implied by the equations, the estimator or the data handling.

Assumption	Kind	Why, and where it breaks
Look $\Delta = 5$ min ahead	chosen	At 1 min, quotes that flicker and snap back look like very fast closing. Cost: closing faster than one step is invisible (§6, step check).
Gap measured from the market's median m	chosen	Removes steady offsets (2024 election: $\approx 3¢$ all month). Breaks when the offset drifts: Super Bowl, $+0.6 \to -0.65¢$.
Flagged one-minute spikes excluded	chosen	One bad quote can dominate least squares. Breaks on real news: FOMC, 11 Sep 12:30 UTC, Polymarket moved first.
Mid-prices are tradable prices	chosen	The fairest single price; real trades pay the spread, and Polymarket's history is only inferred to be mid-prices.
One closing speed κ	hidden	Breaks in the Super Bowl and World Series: fast first, slow afterwards (half-life grows with Δ).
Every 5-minute move is equally noisy	hidden	Near the result and on news, prices jump far more, so those few hours have too much say in the least-squares fit.
Each 5-minute move is independent	hidden	Neighbouring moves share most of their minutes, so they mostly repeat each other; the block bootstrap keeps the error bars from being too narrow.
Symmetric band	hidden	Same c whichever site is higher; fees differ by site and by price (Kalshi $\propto p(1-p)$).
± 30 s matches are simultaneous	hidden	Quiet markets carry stale quotes (holes up to 14 h, papal conclave), which biases $\hat{c}$ and $\hat{\kappa}$ low (§5).
Parameters constant over a market's life	hidden	The last days before resolution are more volatile; the test window sits there.
Both contracts settle on the same outcome	hidden	The arbitrage is risk-free only then. 2024 election: "Trump *or another Republican*" vs "Trump".

the model is

$$x_{t+\Delta} - x_t = -\kappa\, e_c(x_t) + \varepsilon_t, \qquad \mathbb{E}[\varepsilon_t] = 0,\ \mathrm{Var}(\varepsilon_t) = \sigma^2, \quad (3)$$

with $c \geq 0$ the band half-width (cents), $\kappa \in (0,1]$ the share of the excess removed per step, and σ the noise per step (cents). Ignoring noise, the excess obeys $e_{t+\Delta} = (1-\kappa)\, e_t$ and decays geometrically, so it halves after

$$t_{1/2} = \Delta \frac{\ln 2}{-\ln(1-\kappa)} \text{ min.} \quad (4)$$

For small κ, $t_{1/2} \approx 3.5/\kappa$ minutes.

Data pipeline. To make the rule computable we need a gap series sampled on a common clock. We download one-minute prices from both platforms for nine events, three each in politics, sports and culture. Each series stops before the earlier of the two closing times, because at closing Kalshi's book goes to bid 0/ask 1 and its midpoint becomes a misleading 0.50. Kalshi minutes with a spread above $10¢$ are dropped. Each Kalshi minute is paired with the nearest unused Polymarket point within ± 30 s; nothing is resampled or forward-filled, since filling copies old prices forward and invents a lag between the sites. Points more than five median absolute deviations from the local median gap (within ± 15 min, floor $0.5¢$) are flagged.

4 Estimation and uncertainty

Estimator. From every pair of rows exactly Δ apart (each minute a possible start, flagged rows excluded) we form $(x_i,\ \delta_i = x_{i+\Delta} - x_i)$. For fixed c, (3) is a regression through the origin, so

$$\hat{\kappa}(c) = -\frac{\sum_i e_c(x_i)\,\delta_i}{\sum_i e_c(x_i)^2}, \qquad \mathrm{SSE}(c) = \sum_i \big(\delta_i + \hat{\kappa}(c)\, e_c(x_i)\big)^2. \quad (5)$$

We profile over $c \in \{0, 0.1, \ldots, 10\}¢$ and take $\hat{c} = \arg\min_c \mathrm{SSE}(c)$, allowing only bands with at least 300 pairs beyond them, so that no handful of extreme minutes decides c. Then $\hat{\kappa} = \hat{\kappa}(\hat{c})$ and $\hat{\sigma}^2 = \mathrm{SSE}(\hat{c})/n$. A grid is exact here: for fixed c the problem is linear, so nothing can get stuck in a local minimum.

Uncertainty. Neighbouring minutes are strongly dependent, and overlapping pairs share data, so ordinary standard errors would be far too narrow. We use a *block bootstrap*: the pairs are cut into 1-hour blocks, $B = 1000$ datasets are built by drawing blocks with replacement, and c and κ are refitted on each. A 95% range is the 2.5th–97.5th percentile of the refits; the half-life range is the same percentile of $t_{1/2}(\hat{\kappa}_b^*)$, so no error propagation is needed. Percentiles, unlike ± 1.96 standard errors, respect $c \geq 0$ and the strong skew of $t_{1/2}$. The correlation between $\hat{c}$ and the Kalshi spread is a statement about markets, so its range comes from resampling the nine markets. All randomness uses seed 215.

Results. Table 2 and Figure 2 give every estimate with its range. Bands are small, 0–$1.5¢$, and sports markets close fastest (half-lives of 4–7 min), politics in about half an hour, and the thin Swift market in hours. Simulations (§5) show that in the World Series and NBA Finals $\hat{c}$ is 0.4–$0.5¢$ too low and closing looks about twice as slow as it is, because stale quotes hide fast moves; there, $\hat{c}$ is a lower bound and $t_{1/2}$ an upper bound.

5 Verification

The $c = 0$ limit. With no band, (3) is AR(1), $x_{t+1} = (1-\kappa)x_t + \varepsilon_t$, whose properties are known exactly:

$$\rho_k = (1-\kappa)^k, \quad \mathrm{Var}\, x = \frac{\sigma^2}{1-(1-\kappa)^2}, \quad \kappa_h = 1-(1-\kappa)^h. \quad (6)$$

We simulated 50 AR(1) series of 20,000 minutes and ran the production fitting code on them. It returns all three closed-form values to within 0.3% (Table 3). The same check runs as a unit test in GitHub Actions on every pull request.

Planted bands. AR(1) cannot test the part that finds c, so we planted bands of exactly 0, 1, 2 and $3¢$ in fake series built on the Super Bowl's timeline and noise ($\kappa_5 = 0.3$, 50 series each). The median estimates are 0.0, 0.9, 1.9 and $2.9¢$: the

Table 2. Main fit for each market ($\Delta = 5$ min, gap measured from the market's median). Brackets: 95% block-bootstrap ranges. $\hat{\sigma}$: noise per step. m: median offset, on the half-cent price grid, so its range can be a single value. "<1": some resamples remove the whole excess within one step ($\hat{\kappa} \geq 1$). Cost: half the Kalshi spread plus the taker fee $0.07\,p(1-p)$. Skill: out-of-sample, §6.

Market	pairs	$\hat{c}$ (¢)	$t_{1/2}$ (min)	$\hat{\sigma}$ (¢)	m (¢)	cost (¢)	skill (%)
2024 election	20,616	0.8 [0.0, 3.0]	34 [<1, 74]	0.36 [0.31, 0.41]	−3.20 [−3.25, −3.10]	2.2	1.3 [0.5, 2.2]
Sep 2026 FOMC	9,375	0.0 [0.0, 0.3]	59 [44, 78]	0.46 [0.41, 0.51]	0.00 [0.00, 1.00]	2.0	3.0 [0.5, 4.6]
Oct 2025 shutdown	14,637	0.9 [0.3, 2.2]	36 [19, 55]	0.89 [0.81, 0.98]	0.00 [0.00, 0.50]	2.3	4.4 [2.7, 5.7]
Super Bowl LX	45,145	1.2 [1.0, 2.0]	7.0 [<1, 15]	0.30 [0.26, 0.33]	0.25 [0.20, 0.30]	2.0	−62 [−515, −14]
2025 World Series	21,360	1.3 [0.8, 1.8]	3.6 [1.8, 6.8]	0.73 [0.65, 0.81]	1.00 [1.00, 1.00]	2.4	13.4 [12.1, 14.6]
2025 NBA Finals	32,438	1.3 [0.8, 2.3]	5.4 [<1, 9.7]	0.41 [0.36, 0.47]	−0.50 [−0.50, −0.50]	1.9	12.0 [4.3, 19.6]
Papal conclave	1,290	0.6 [0.0, 0.7]	3.9 [<1, 37]	0.22 [0.16, 0.27]	0.05 [0.00, 0.15]	0.6	2.5 [1.8, 3.1]
2025 Oscars	1,201	1.0 [0.0, 2.8]	18 [6.2, 36]	1.19 [0.96, 1.36]	−1.00 [−1.00, −0.50]	2.3	6.4 [2.5, 9.5]
Swift engagement	8,315	1.5 [0.0, 8.9]	173 [17, 411]	0.61 [0.52, 0.70]	0.50 [0.00, 1.00]	3.0	0.5 [−0.0, 1.0]

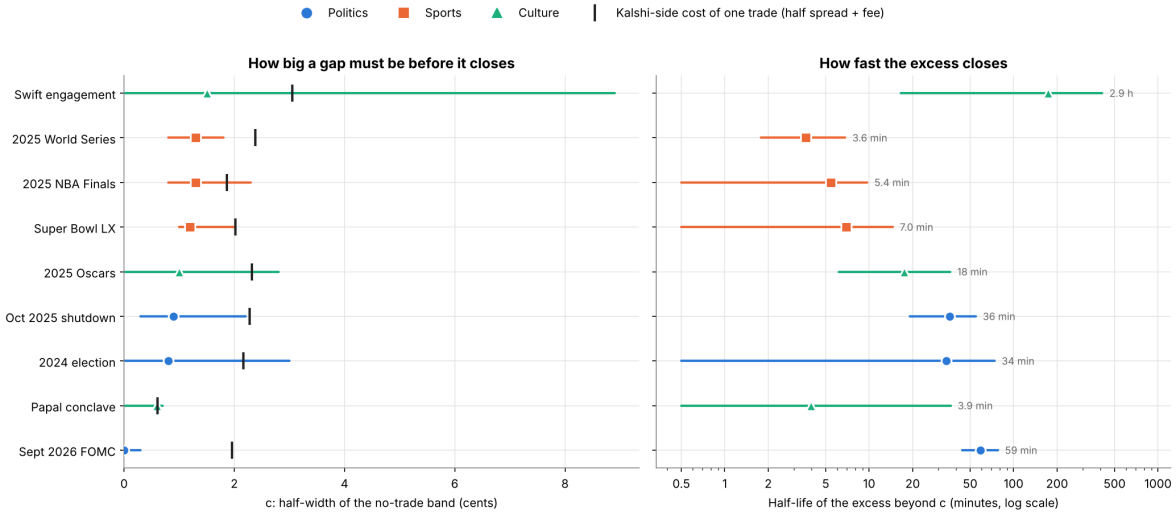


Figure 2. Fitted band $\hat{c}$ (left; black ticks: Kalshi-side cost of one trade) and half-life of the excess (right, log scale), with 95% block-bootstrap ranges. A bar reaching 0.5 min means some resamples remove the whole excess within one 5-minute step.

Table 3. Verification in the $c = 0$ limit: 50 simulated AR(1) series, $\kappa = 0.05$ per minute, $\sigma = 0.3$¢.

Quantity	Closed form (6)	Code
$\hat{\kappa}$, 1-min step	$\kappa = 0.0500$	0.0499
$\hat{\kappa}$, 5-min step	$1-(1-\kappa)^5 = 0.2262$	0.2255
Spread of x	$\sigma/\sqrt{1-(1-\kappa)^2} = 0.961$¢	0.961¢

method never invents a band and recovers real ones slightly low. Measuring the gap from a trailing 24-hour median instead returns 0.0, 0.2, 0.0, 0.0¢; the moving center follows the gap's wandering inside the band and erases it, which is why we use one fixed median per market (Figure 3b).

Realistic fake data, market by market. Finally, for each market we built 100 fake series with *that market's* fitted c and κ, its holes in the timeline, half-cent rounding, one-minute flicker (sized from the negative autocovariance of its one-minute changes, $\mathrm{cov}(\delta_t, \delta_{t+1}) = -\tau^2$) and stale quotes (tuned to its share of unchanged minutes), and refitted each with its own bootstrap (Figure 3a). The 95% ranges contain the planted c in 91–96% of runs for six markets, close to nominal, but in 0–6% for the World Series, NBA Finals and papal conclave, where $\hat{c}$ comes out 0.2–0.5¢ low. This is the source of the bias stated in §4.

6 Validation and a baseline

Baseline. The baseline predicts *no change* in the next five minutes, a random walk. It has no parameters, so nothing is fitted and nothing can leak from the test data. It is also the standard benchmark for prices: if markets were perfectly efficient, future changes would be unpredictable and "no change" would be the best forecast, so beating it is direct evidence that gaps are pulled back. Plain AR(1) serves as a second baseline that isolates what the band adds.

Metric. Each market is split in time: c, κ and m are fitted on its first 70%, and the last 30% is the test set. For the N test pairs exactly 5 minutes apart, the target is the true change $y_i = x_{t_i+5} - x_{t_i}$ and each model predicts $\hat{y}_i$ from x_{t_i} alone: $\hat{y}_i = -\hat{\kappa}\, e_{\hat{c}}(x_{t_i})$ (band), $-\hat{\kappa}_0 x_{t_i}$ (AR(1)) or 0 (baseline). We report

$$\text{skill} = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{RW}}}, \qquad \text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2, \tag{7}$$

with 95% ranges from resampling 1-hour blocks of the test period. Most 5-minute changes are noise that nobody can predict, so a few percent is a real gain.

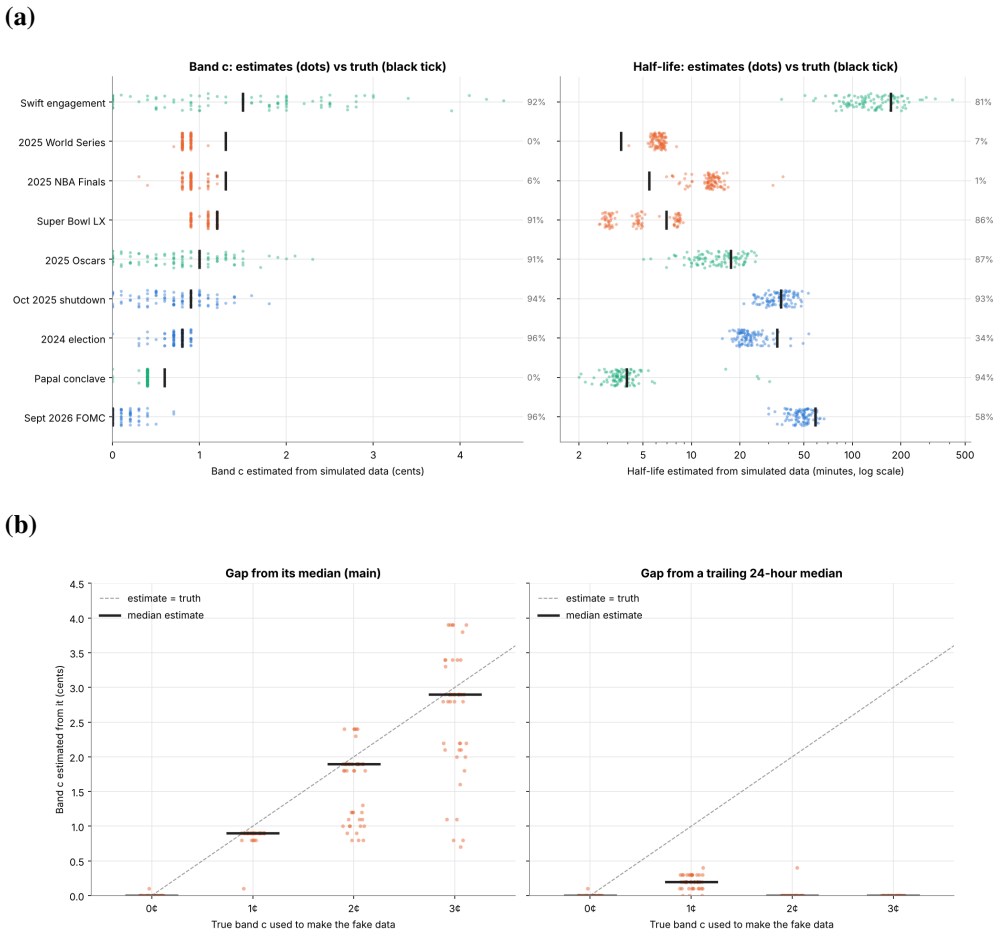


Figure 3. Verification by simulation. (a) For each market, 100 fake series built with its own fitted c and κ, timeline, rounding, flicker and stale quotes: dots are the estimates of c (left) and of the half-life (right), black ticks the planted values, and the numbers at the right the share of 95% ranges that contain the planted value. (b) Bands of exactly 0–3¢ planted on the Super Bowl template, 50 series each; the fixed median (left) recovers them, a trailing 24-hour median (right) erases them.

Table 4. Held-out markets, same pipeline and score as Table 2. "Same": c, κ fitted on the market's first 70%. "Pooled": median c and κ of the original nine, nothing fitted. "—": fewer than 300 pairs beyond any c.

Market	pairs	$\hat{c}$ (¢)	$t_{1/2}$ (min)	cost (¢)	skill: same (%)	skill: pooled (%)	skill: AR(1) (%)
Biden resigns	67	—	—	1.6	—	15.9 [8.4, 26.7]	10.9 [7.5, 27.3]
Texas Senate	1,097	2.4 [0.0, 2.9]	7.4 [2.7, 86]	1.8	—	7.0 [−6.3, 10.0]	1.2 [0.9, 1.9]
NYC mayor	9,173	0.4 [0.0, 2.4]	17 [<1, 28]	1.9	6.4 [4.7, 7.5]	7.0 [4.9, 8.7]	6.4 [4.7, 7.5]
Spain World Cup	4,126	0.0 [0.0, 0.0]	98 [71, 147]	1.7	−79 [−144, −45]	−6.5 [−11.7, −3.7]	−2.9 [−8.7, 0.3]
Stanley Cup	5,081	1.8 [1.3, 1.9]	<1	2.3	−36 [−371, 8.5]	7.3 [−8.9, 10.3]	−64 [−544, −11]
Masters	7,975	0.8 [0.7, 0.9]	<1	1.3	0.6 [0.6, 0.6]	9.1 [7.7, 10.0]	0.6 [0.6, 0.6]
Oscars Best Picture	9,795	0.0 [0.0, 2.0]	213 [25, 323]	1.8	−0.5 [−3.1, 1.5]	−43 [−75, −23]	−0.5 [−3.1, 1.5]
Oscars Best Actor	13,867	2.5 [1.8, 3.1]	2.1 [<1, 6.5]	2.2	−9.8 [−19.4, −2.5]	−1.0 [−5.2, 2.5]	−2.8 [−7.7, 1.1]
Oscars Best Actress	3,393	0.9 [0.0, 0.9]	8.0 [4.3, 106]	0.8	−10.6 [−39.9, −2.7]	0.7 [−0.2, 2.9]	−7.7 [−24.9, −1.6]

Later data. The band model beats the random walk in 7 of 9 markets, with the whole 95% range above zero, by 1–13% (Figure 4a); it beats plain AR(1) in three (FOMC, shutdown, World Series). It fails badly in the Super Bowl (−62%), whose platform offset drifted between training and test: measuring its gap from a trailing median instead gives +16.5% (Figure 4b), confirming the drift rather than a failure of the band rule itself.

Held-out markets. Nine further markets (PR #3) were used for no fit and no choice of setting. We ran the unchanged pipeline two ways: fitting c, κ on each new market's first 70%, and with no fitting at all, using the median $c = 1.0$¢ and $\kappa = 0.18$ of the original nine (Table 4, Figure 6). The pull is clean, but the markets were not chosen by a written rule, and several are poor tests: the Spain World Cup data are four months before the tournament (prices flat in 97–99.8% of minutes); Biden-resign and the Masters are long shots near 6¢ (Biden has only 67 usable pairs; the Masters gap is exactly zero in 53% of minutes); Best Actress is near-certain and barely moves; and NYC mayor compares "nominee" with "primary winner". Fitted per market, the band beats the random walk

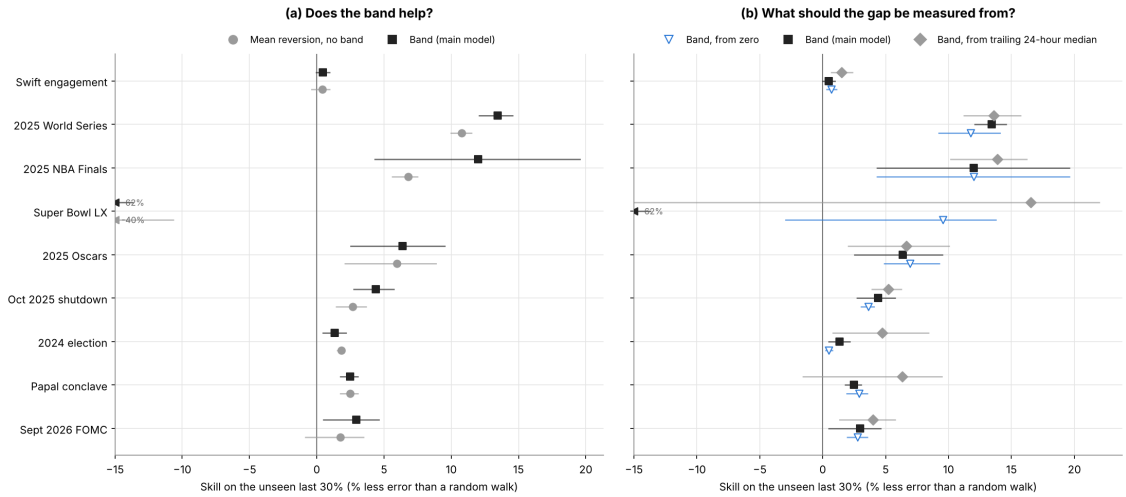


Figure 4. Out-of-sample skill on the last 30% of each market: % less squared error than the random walk; bars are 95% block-bootstrap ranges. (a) Band model vs plain AR(1). (b) The band with the gap measured from zero, from the training median (main) or from a trailing 24-hour median.

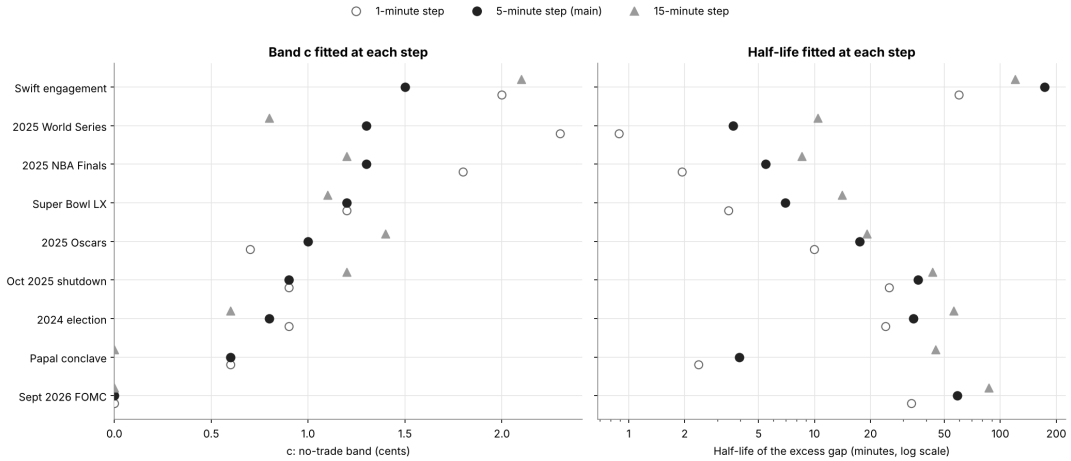


Figure 5. Time-step check: band c (left) and half-life (right, log scale) fitted with 1-minute (open circles), 5-minute (filled) and 15-minute (triangles) steps. With one closing speed the three markers would coincide. A much shorter 1-minute half-life is quote flicker; one that keeps growing from 5 to 15 minutes means two-stage closing.

in 2 of 7 fittable markets; with pooled parameters, skill is positive in 6 of 9 (clearly so in 3); plain AR(1) wins in 4 of 9. The test therefore does not confirm that the band carries over to new markets, but with these markets it cannot show that it fails either.

An independent measurement. The model never sees costs. If c is a trading cost, it should rise with the measured Kalshi-side cost, and it does: Spearman $\rho = +0.77$ (95% range $+0.16$ to $+1.00$) across the nine markets, although only $+0.25$ on the held-out set. The fitted $\hat{c}$ sits *below* the quoted cost in 8 of 9 markets. That cost assumes a trader who crosses the spread and pays the full taker fee; arbitrageurs who post limit orders pay neither, and stale quotes bias $\hat{c}$ low (§5). So c tracks cost without equalling the quoted taker cost.

One closing speed. If a single κ were right, the half-life in minutes would not depend on the step. It is stable at 1, 5 and 15 minutes for the politics markets and the Oscars, but keeps growing in the Super Bowl (3.5, 7.0, 14 min) and World Series (0.9, 3.6, 10 min): gaps there close in two stages (Figure 5). Consistently, real gaps stay correlated far longer than (6) predicts (2024 election: 0.64 after two hours, against 0.29).

7 What changed after M2

TF feedback. "Justify your choice of baseline model": taken up in §6, which explains why the random walk is the benchmark and adds AR(1) as a second baseline. "Explain how you evaluated the model": taken up in §6, which gives the time split, target, MSE, skill, bootstrap ranges and the held-out test. "Slide text too small": applies to the presentation; here every figure is set at two-thirds to full text width, its explanation is in the caption, and every number is also tabulated.

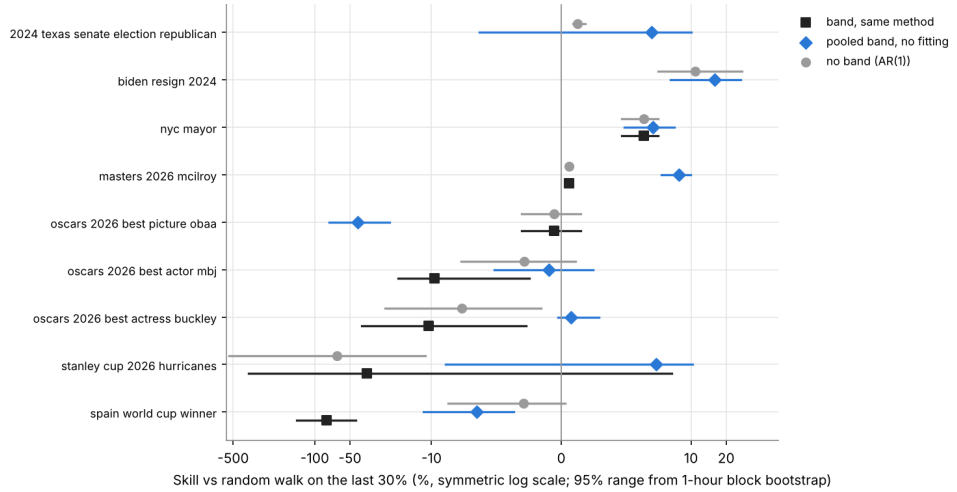


Figure 6. Held-out markets: skill on the last 30% of each market against the random walk (symmetric log scale; bars are 95% block-bootstrap ranges) for the band fitted on that market's first 70% (squares), the pooled band of the original nine with nothing fitted (diamonds), and AR(1) (circles). Same numbers as Table 4.

Changes from our M2 plan.

- *5-minute data only on Polymarket* → its CLOB endpoint gives 1-minute data; $\Delta = 5$ is now a choice, tested at 1, 5 and 15 min.
- *52¢ pilot gap ignored as an error* → traced to settlement (bid 0/ask 1) and fixed at the source.
- *Gap from zero* → from each market's median, because of steady offsets.
- *One pooled model* → fitted per market (bands 0–1.5¢, half-lives minutes to hours); pooling kept as the no-fit test.
- *Hourly forward-filled pilot* → ±30 s matching with no filling.
- *Planned* error bars, $c = 0$ check, baseline, fee schedule, and repository: all done; PR #5 adds the `src/` package, `pyproject.toml`, `uv.lock` and pytest tests run by GitHub Actions.

8 What this does not show

What the data cannot decide. Over long horizons a slowly drifting offset and a gap wandering inside the band look the same, so c is not identified when the offset moves. Three markets per topic cannot establish differences between topics. Without Polymarket's historical spreads we cannot measure the full round-trip cost, nor how much of a steady offset is a premium for settlement risk.

What the model leaves out. A second, slower closing speed; a moving offset; news jumps; asymmetric costs; and the cost of capital locked until resolution, which should make c larger for events far in the future.

What another month would buy. A time-varying κ_t, fitted in sliding windows or as a slowly evolving state in a state-space model, with the window length trading responsiveness against noise. Jumps in the noise, $\varepsilon_t + B_t J_t$ with $B_t \sim \text{Bernoulli}(\lambda)$, fitted with c and κ by maximum likelihood, which would replace hand-flagging spikes and measure closing right after news. A few days of live order-book recording on both sites to measure true costs. And 10–20 further markets chosen by a rule fixed in advance (≥5,000 pairs, median price 10–90¢, data up to market close, identical resolution wording), which would make the held-out test fair.

9 Contributions

Alexander Meylikhov: one-event pilot (`pilot_pull.py`, `FOMC.ipynb`, committed by Ravi in `50ec973`); PR #1, the nine-event pull notebook and `results/`; PR #2, the corrected pull (settlement artifacts, units, 1-minute data, ±30 s matching, outlier flags), `fit_kappa.ipynb` (fit, bootstrap, all checks, out-of-sample test) and README; Part D (ranges for σ, m, ρ; held-out test); PR #5, the `src/mtp_gaps` package, `pyproject.toml`, `uv.lock`, pytest tests and CI; merged PR #3; M2 write-up, checkpoint-2 presentation and this report.

Ravi Tripathi: created the repository and its initial commits (`e32a951–d4eda70`); reviewed and merged PR #5.

Ved Pant: reviewed and merged PRs #1 and #2; PR #3, the nine held-out markets in `verification_data/`.